

Formalizing the homotopy extension property in LEAN

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Summary

This thesis is about the formalisation of the homotopy extension property in the formal language of the proof assistant Lean.

After a short introduction section 1.1, the chapter section 2 gives an overview of the mathematics behind the formalisation. We see the definition, in equivalent phrasings, the retraction criterion, which is used to show a pair of topological spaces satisfies the homotopy extension property and have two statements about the conversation of the property. Afterwards we encounter the important instance, that relative CW-complexes satisfy the homotopy extension property. The formalisation continues up to the proof of this result for finite dimensional complexes.

The following chapter is about Lean and mathlib (cref `ToDo`). Apart from a general short overview I highlight three concepts, relevant for this project. Namely the difference between sets and subtypes, structures and junk values. I briefly mention the library of formalised mathematics, called mathlib.

The chapter (cref `ToDo`) makes up a main part of this thesis. Based on short sequences of my formalisation I explain and judge the decisions made along the way. Main theorems from mathlib, which some proofs are build on, are introduced.

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1 Introduction

In this work, if not specified in any other way, every space has a topology and every map is a continuous map.

1.1 Motivation

The goal of this thesis is to formalize the *homotopy extension property (HEP)* in Lean. I started with the proof of the *retraction criterion* and worked towards the result that every relative CW complex satisfies the HEP. Our motivation for studying this property is that it is needed for two major theorems in algebraic topology. The first is the *cellular approximation theorem*.

Theorem 1.1 (Cellular Approximation Theorem, [Hat02, Theorem 4.8]). *Every map $f: X \rightarrow Y$ of CW complexes is homotopic to a cellular map. If f is already cellular on a subcomplex $A \subset X$, the homotopy may be taken to be stationary on A .*

Here a map is called *cellular* if it sends the n -skeleton of X to the n -skeleton of Y for all $n \in \mathbb{N}$.

Equipping S^k with the CW structure consisting of one 0-cell and one k -cell, it has no cells in dimensions $0 < n < k$. Hence any map $S^n \rightarrow S^k$ with $n < k$ is homotopic to one landing in the n -skeleton, which is a single point, and is therefore nullhomotopic. This yields the corollary $\pi_n(S^k) = 0$ for all $n < k$.

Secondly, the result that relative CW complexes satisfy the HEP is one of the main steps towards the proof of *Whitehead's theorem*.

Theorem 1.2 (Whitehead's Theorem, [Hat02, Theorem 4.5]). *If a map $f: X \rightarrow Y$ between connected CW complexes induces isomorphisms $f_*: \pi_n(X) \rightarrow \pi_n(Y)$ for all n , then f is a homotopy equivalence. In case f is the inclusion of a subcomplex $X \hookrightarrow Y$, the conclusion is stronger: X is a deformation retract of Y .*

Whitehead's theorem states that, in order to prove that two connected CW complexes are homotopy equivalent, it suffices to exhibit a map between them that induces isomorphisms on all homotopy groups, i.e. a weak homotopy equivalence. Showing that two spaces are homotopy equivalent is in general a difficult task, and Whitehead's theorem provides a powerful tool for such problems.

2 Homotopy Extension Property

Our formalisation is not based on a single source. The overall structure follows the lecture notes of the course Topology I, taught by Markus Hausmann at the University of Bonn in the winter term 2025/26. These notes are not publicly available; the chapter on the homotopy extension property closely follows the (German) lecture notes Algebraische Topologie by F. Waldhausen (Bielefeld) [], which we cite instead wherever a specific argument is taken from them. Throughout, all maps between topological spaces are assumed to be continuous, and we write $I := [0, 1]$ for the unit interval.

2.1 The Homotopy Extension Property

There are several equivalent ways to define the homotopy extension property, and this section We begin with the definition that is closest to the one we formalised in Lean.

– Question: nicht genau so zitiert, aber es gibt ähnliche definition in der Quelle, dann schribbe ich die Quelle dran?

Definition 2.1. ?? A pair of topological spaces (X, A) with $A \subseteq X$ has the *homotopy extension property (HEP)* if for every topological space Y , every map $f: X \rightarrow Y$, and every homotopy $H: A \times I \rightarrow Y$ satisfying $H(a, 0) = f(a)$ for all $a \in A$, there exists a homotopy $H': X \times [0, 1] \rightarrow Y$ such that

$$\begin{aligned} H'(x, 0) &= f(x) && \text{for all } x \in X, \\ H'(a, t) &= H(a, t) && \text{for all } (a, t) \in A \times [0, 1]. \end{aligned}$$

The two extreme cases are immediate, but we record them separately because we will use them later as the base case of an induction.

Lemma 2.2. *For every topological space X , both (X, X) and $(X, \emptyset)$ have the HEP.*

Proof. For (X, X) , the homotopy H is already defined on all of $X \times I$ and satisfies $H(x, 0) = f(x)$, so we may take $H' := H$.

For $(X, \emptyset)$, the second condition of Definition 2.1 is vacuous, so the constant homotopy $H'(x, t) := f(x)$ has the required properties. $\square$

Remark 2.3. The HEP can equivalently be phrased as follows: (X, A) has the HEP if and only if every map defined on the pushout $X \cup_A (A \times I)$ extends to a map on $X \times I$. Here the pushout is taken along the inclusion $A \hookrightarrow X$ and the map $A \rightarrow A \times I, a \mapsto (a, 0)$. This is in general, *not* the same as requiring that every map defined on the subspace $X \times \{0\} \cup (A \times I) \subset X \times I$ extends to $X \times I$, because the canonical map

$$X \cup_A (A \times I) \rightarrow X \times \{0\} \cup (A \times I)$$

need not be a necessarily a homeomorphism. In general, it is only a continuous bijection. If A is closed in X , however, both $X \times \{0\}$ and $A \times I$ are closed in $X \times I$, the canonical map is a closed map and hence a homeomorphism, and the two formulations agree. This is the reason why the closedness assumption appears in exactly one direction of lemma 2.4 below.

Lemma 2.4. *Let A be a subset of a topological space X and consider the following two statements*

(i) *The pair (X, A) has the HEP*

(ii) *For every topological space Y and every continuous map $g: (X \times \{0\}) \cup (A \times I) \rightarrow Y$ there exists an extension of g to a map $G: X \times I \rightarrow Y$*

Then (i) implies (ii). If in addition A is closed in X , then also (ii) implies (i).

Proof. (i) $\Rightarrow$ (ii). Let $g: X \times \{0\} \cup A \times I \rightarrow Y$ be a map. Define

$$f: X \rightarrow Y, x \mapsto g(x, 0) \quad \text{and} \quad H: A \times [0, 1] \rightarrow Y, (x, t) \mapsto g(x, t).$$

Both are continuous, being composites of g with the continuous maps $x \mapsto (x, 0)$ and $A \times I \hookrightarrow (X \times \{0\}) \cup (A \times I)$. They are compatible in the sense of definition 2.1, since $H(a, 0) = g(a, 0) = f(a)$ for all $a \in A$. Applying the HEP of (X, A) yields a homotopy $H': X \times I \rightarrow Y$ with $H'(x, 0) = f(x) = g(x, 0)$ and $H'|_{A \times I} = H = g|_{A \times I}$. Hence $G := H'$ restricts to g on all of $(X \times \{0\}) \cup (A \times I)$.

(ii) $\Rightarrow$ (i), assuming A closed. Let $f: X \rightarrow Y$ and $H: A \times I \rightarrow Y$ with $H(a, 0) = f(a)$ for all $a \in A$. We define

$$g: X \times \{0\} \cup A \times [0, 1] \rightarrow Y, \quad (x, t) \mapsto \begin{cases} f(x), & t = 0, \\ H(x, t), & \text{otherwise} \end{cases}$$

The map g is well defined: the two cases overlap precisely on $A \times \{0\}$, where $H(a, 0) = f(a)$ by assumption. For continuity we use that A is closed in X : then $A \times I$ is closed in $X \times I$, and so is $X \times \{0\}$. Both are therefore closed in the subspace $(X \times \{0\}) \cup (A \times I)$, which they cover, and the restrictions of g to them are continuous. By assumption there is an extension $G: X \times I$ of g , and $H' := G$ satisfies both conditions of definition 2.1 by construction. $\square$

Definition 2.5. Let X be a topological space and $A \subset X$ a subset with inclusion $\iota: A \hookrightarrow X$. A *retraction* of X onto A is a map $r: X \rightarrow A$ with $r \circ \iota = id_A$. If such an r exists, we call A a *retract* of X .

Lemma 2.6. *Let A be a subset of a topological space X . The following are equivalent:*

(i) For every topological space Y and every $g: A \rightarrow Y$ there exists an extension $G: X \rightarrow Y$

(ii) A is a retract of X .

Proof. (i) $\Rightarrow$ (ii). Apply (i) to the special case $Y := A$ and $g := id_A$. The resulting extension G restricts to the identity on A and is therefore a retraction.

(ii) $\Rightarrow$ (i). Let $g: A \rightarrow Y$ be a map and let $r: X \rightarrow A$ be a retraction. Then $G := g \circ r$ is continuous, and for $a \in A$ we have $G(a) = g(r(a)) = g(a)$, so G extends g . $\square$

Combining the two lemmas gives the *retraction criterion*, which is the tool we use in several of the following proofs: it replaces the quantification over all spaces Y and all maps into them by the existence of a single map. In the lecture notes the criterion is stated only for closed subspaces $A \subseteq X$. Hatcher [Hat02] states it without any such assumption, but his proof of the implication from the retraction to the HEP again requires A to be closed; for the general case he refers to the appendix, where a detailed proof following an argument of [Str68] is given. Our formulation follows the same pattern as lemma 2.4: we impose closedness only on the implication that actually needs it.

For our purposes this restriction is harmless, since in all cases we consider the closedness of the subspace is available at no additional cost. –ToDo : Ausblick: verallgemeinern

Proposition 2.7 (Retraction Criterion). *Let X be a topological space and $A \subseteq X$ a subspace. Consider the following two statements:*

(i) The pair (X, A) has the HEP

(ii) the subspace $X \times \{0\} \cup A \times I$ is a retract of $X \times I$

Then (i) implies (ii). If in addition A is closed in X , then (ii) also implies (i).

Proof. (i) $\Rightarrow$ (ii). By lemma 2.4, every map defined on $(X \times \{0\}) \cup (A \times I)$ extends to $X \times I$. Applying lemma 2.6 to the subspace $(X \times \{0\}) \cup (A \times I) \subseteq X \times I$ yields the retraction.

(ii) $\Rightarrow$ (i), assuming A closed. By lemma 2.6, applied to the same subspace, every map defined on $(X \times \{0\}) \cup (A \times I)$ extends to $X \times I$. Since A is closed in X , the second direction of lemma 2.4 applies and yields the HEP for (X, A) . $\square$

The remaining results of this section are permanence properties: they allow us to build new pairs with the HEP out of known ones. The first one is used in ?? to transfer the HEP from balls to cubes, the second one is the inductive step in the proof for relative CW-complexes.

Proposition 2.8. *Let (X, A) and (Y, B) be pairs of topological spaces with B closed in Y , and let $\varphi: X \rightarrow Y$ be a homeomorphism with $\varphi(A) = B$. If (X, A) has the HEP, then so does (Y, B) .*

Proof. Applying proposition 2.7 to (X, A) gives a retraction $r_X: X \times I \rightarrow (X \times \{0\}) \cup (A \times I)$. Let $pr_X: X \times I \rightarrow X$ and $pr_I: X \times I \rightarrow I$ denote the projections onto the first and second component, respectively, and define

$$r_Y: Y \times I \longrightarrow (Y \times \{0\}) \cup (B \times I)$$

$$(y, t) \longmapsto \left(\varphi \circ pr_X \circ r_X(\varphi^{-1}(y), t), pr_I \circ r_X(\varphi^{-1}(y), t) \right).$$

We show that r_Y is a retraction. By proposition 2.7 this yields the HEP for the pair (Y, B) , since B is closed in Y . The map r_Y

- *is continuous*, since both component functions are composites of continuous maps.
- *takes values in $(Y \times \{0\}) \cup (B \times I)$* : let $(y, t) \in Y \times I$. The point $r_X(\varphi^{-1}(y), t)$ lies in the image of r_X , which is $(X \times \{0\}) \cup (A \times I)$. Hence either its second component is 0, in which case the second component of $r_Y(y, t)$ is 0 as well, or its first component lies in A , in which case the first component of $r_Y(y, t)$ lies in $\varphi(A) = B$.
- *restricts to the identity on $(Y \times \{0\}) \cup (B \times I)$* : let (y, t) be a point of this subspace. Since φ^{-1} maps Y to X and B to A , the point $(\varphi^{-1}(y), t)$ lies in $(X \times \{0\}) \cup (A \times I)$, on which r_X is the identity. Therefore $r_X(\varphi^{-1}(y), t) = (\varphi^{-1}(y), t)$, and applying φ to the first component gives $r_Y(y, t) = (y, t)$.

□

Proposition 2.9. *Let $C \subseteq B \subseteq X$ be topological spaces such that both (X, B) and (B, C) have the HEP. Then (X, C) has the HEP.*

Proof. Let Z be a topological space, let $f: X \rightarrow Z$ be a map and let $H: C \times I \rightarrow Z$ be a homotopy with $H(c, 0) = f(c)$ for all $c \in C$. Applying the HEP of (B, C) to the restriction $f|_B$ and to H yields a homotopy $H': B \times I \rightarrow Z$ with

$$H'(b, 0) = f(b) \quad \text{for all } b \in B \quad \text{and} \quad H'|_{C \times I} = H.$$

In particular $H'(b, 0) = f(b)$, so we may apply the HEP of (X, B) to f and H' and obtain a homotopy $H'': X \times I \rightarrow Z$ with $H''(x, 0) = f(x)$ for all $x \in X$ and $H''|_{B \times I} = H'$. The first condition of definition 2.1 is thus satisfied, and since $C \subseteq B$ we also get $H''|_{C \times I} = H'|_{C \times I} = H$. □

2.2 HEP for relative CW-complexes

To proof the *HEP* for relative CW-complexes, we generalize the pair of spaces in three steps proving the HEP at each stage:

1. Every closed ball relative to its boundary has the HEP.
2. For every relative CW-complex, two consecutive skeleta form a pair with the HEP, that is, (X_{n+1}, X_n) has the HEP for all n .
3. Every relative CW-complex (X, C) has the HEP.

Before turning to the proofs we introduce the definitions and notations we use for CW-complexes.

The following definition of relative CW complexes is a modified version of the definition of CW-complexes given in the bachelor thesis of Hannah Scholz [Sch24]. She formalised the historical definition of CW-complexes, presented by Whithead in [Whi18], and this formalisation is now available in mathlib as *Classical CW-Complex*. We follow her definition because it provides exactly the structures our formalised proofs rely on: explicit characteristic maps together with the weak topology, rather than an inductive description via attaching cells.

Definition 2.10. We write $D^n := \{x \in \mathbb{R}^n \mid \|x\| \leq 1\}$ n -dimensional closed ball. Its boundary $\partial D^n = S^{n-1} := \{x \in \mathbb{R}^n \mid \|x\| = 1\}$ is the $(n - 1)$ -dimensional sphere.

Definition 2.11. Let X be a Hausdorff space and $C \subset X$ a closed subspace. A *relative CW-complex* on (X, C) consists of a family of indexing sets $(I_n)_{n \in \mathbb{N}}$ together with a family of maps $(Q_i^n : D_i^n \rightarrow X)_{n \geq 0, i \in I_n}$ such that:

1. $Q_i^n|_{\mathring{D}_i^n} : \mathring{D}_i^n \rightarrow Q_i^n(\mathring{D}_i^n)$ is a homeomorphism. We call $e_i^n := Q_i^n(\mathring{D}_i^n)$ an *(open) n -cell* (or a cell of dimension n) and $\bar{e}_i^n := Q_i^n(D_i^n)$ a *closed n -cell*.
2. For all $n, m \in \mathbb{N}$, $i \in I_n$ and $j \in I_m$ with $(n, i) \neq (m, j)$ the cells e_i^n and e_j^m are disjoint.
3. For all $n \in \mathbb{N}$ and $i \in I_n$, the set $Q_i^n(\partial D_i^n)$ is contained in the union of C and a finitely many closed cells of dimension less than n .
4. A subset $S \subseteq X$ is closed if and only if $S \cap C$ is closed and $S \cap \bar{e}_i^n$ is closed for all $n \in \mathbb{N}$ and $i \in I_n$.
5. $C \cup \bigcup_{n \geq 0} \bigcup_{i \in I_n} Q_i^n(D_i^n) = X$.

We call Q_i^n a *characteristic map*, refer to C as the *base* of the complex and call $\partial e_i^n := Q_i^n(\partial D_i^n)$ the *frontier of the n -cell* for any i and n . For $-1 \leq n \leq \infty$ we define the n -skeleton of X as

$$X_n := C \cup \bigcup_{m \leq n} \bigcup_{i \in I_m} \bar{e}_i^m.$$

Note that $X_{-1} = C$ and $X_\infty = X$. Moreover, every skeleton is closed in X . This follows from condition (4), since the intersection of X_n with C and with every closed cell is closed. We use this below whenever we apply the converse direction of proposition 2.7.

Definition 2.12. If $C = \emptyset$, we call $(X, \emptyset)$, or simply X , an *(absolute) CW-complex*.

Definition 2.13. A relative CW complex (X, C) is called *finite dimensional* if there exists an $n \in \mathbb{N}$ such that $I_m = \emptyset$ for all $m \geq n$.

2.2.1 HEP for balls

Theorem 2.14. *For every $m \in \mathbb{N}$, the pair $(D^m, \partial D^m)$ has the HEP.*

There are two different attempts to prove this. One argues visually via the retraction criterion, projection the cylinder $D^m \times I$ from a point above it onto $(D^m \times \{0\}) \cup (\partial D^m \times I)$. The other constructs the extended homotopy explicitly. We give the constructive proof, since it was the easier one to formalise in Lean.

Proof. Let Y be a topological space, let $f: D^m \rightarrow Y$ be a map and let $H: \partial D^m \times I \rightarrow Y$ be a homotopy with $H(a, 0) = f(a)$ for all $a \in \partial D^m$. We assemble these into a map $h: D_2^m \rightarrow Y$, where D_2^m denotes the closed ball of radius 2 in $\mathbb{R}^m$, by setting

$$h(x) = \begin{cases} f(x) & \|x\| \leq 1 \\ H\left(\frac{x}{\|x\|}, \|x\| - 1\right), & \|x\| \geq 1 \end{cases}$$

The two cases agree on the sphere $\|x\| = 1$, where the second one gives $H(x, 0) = f(x)$, and both domains are closed in D_2^m , so h is continuous by the pasting lemma. Using h we define

$$H': D^m \times I \rightarrow Y, \quad H'(x, t) = h((1+t)x),$$

which is well defined and continuous, and which satisfies $H'(x, 0) = h(x) = f(x)$ for all $x \in D^m$ as well as $H'(a, t) = H(a, t)$ for all $a \in \partial D^m$ and $t \in I$, since $\|(1+t)a\| = 1+t \geq 1$. $\square$

For $m = 0$ the sphere ∂D^0 is empty and the second case of the definition of h never occurs. In Lean this requires no special treatment, since division by zero is defined to be zero: the term $x/\|x\|$ is total and the case distinction goes through unchanged.

Since the definition of CW-complexes in mathlib uses n -dimensional cubes rather than closed balls as the domain of the characteristic maps, we restate the HEP for cubes relative to their boundary. – ToDo: Question: eigentlich nicht ist der domain nicht der cube, sondern es ist ein partial homeomorphismus mit source cube. Muss man da präzieser sein, oder reicht das, weil es hier nur um die Motivation geht?

Corollary 2.15. *For every $n \in \mathbb{N}$, the pair $([-1, 1]^n, \partial[-1, 1]^n)$ has the HEP.*

Proof. The ball D^n is homeomorphic to the cube $[-1, 1]^n$ via radial scaling, which maps each ray from the origin to itself and takes ∂D^n to $\partial[-1, 1]^n$. This maps the sphere to the boundary of the cube. The claim now follows from theorem 2.14 together with proposition 2.8. $\square$

2.2.2 HEP for consecutive skeletons

In the next step we prove that for every relative CW complex (X, C) and every $m \in \mathbb{N}$, the pair (X_m, X_{m-1}) has the HEP. We do so by applying the retraction criterion, that is, by constructing a retraction

$$r_m^{\text{Skel}}: X_m \times I \rightarrow (X_m \times \{0\}) \cup (X_{m-1} \times I).$$

This map will be obtained from the universal property of the quotient. To use it we need to know that $X_m \times I$ carries the quotient topology, and for this it suffices to know that X_m does, by the following theorem:

Theorem 2.16. *Let Z be a locally compact space and let $g: Y \rightarrow Y'$ be a quotient map. Then*

$$g \times \text{id}_Z: Y \times Z \rightarrow Y' \times Z$$

is a quotient map.

Since the unit interval I is compact, and in particular locally compact, theorem 2.16 applies to $Z := I$ throughout this section. So let us prove that every skeleton indeed is a quotient space: We set $Z_m := C \amalg \coprod_{n \leq m} \coprod_{i \in I_n} D_i^n$ and define

$$P_m: Z_m \rightarrow X_m$$

$$x \mapsto \begin{cases} Q_i^n(x), & x \in D_i^n, \\ x, & x \in C \end{cases}$$

Lemma 2.17. *Let (X, C) be a relative CW-complex. For every $m \in \mathbb{N}$ the map $P_m : Z_m \rightarrow X_m$ is a quotient map .*

Proof. The map P_m is well defined, since the images of the characteristic maps up to dimension m lie in X_m and the base C is contained in every skeleton. It is continuous because its restriction to each summand of the coproduct is, and it is surjective by the definition of X_m .

It remains to show that a subset $S \subseteq X_m$ is closed whenever $P_m^{-1}(S)$ is closed. So let $P_m^{-1}(S)$ be closed. By the definition of the coproduct topology this means that

$$P_m^{-1}(S) \cap C = S \cap C \quad \text{and} \quad P_m^{-1}(S) \cap D_i^n = (Q_i^n)^{-1}(S)$$

are closed for all $n \leq m$ and $i \in I_n$. Since X_m is closed in X , it suffices to show that S is closed in X . By condition 4 of definition 2.11 this amounts to showing that $S \cap C$ and $S \cap \bar{e}_i^n$ are closed for all $n \in \mathbb{N}$ and $i \in I_n$.

$S \cap C$ is closed by the above. For $n \leq m$ we have $S \cap \bar{e}_i^n = Q_i^n((Q_i^n)^{-1}(S))$, which is closed as the image of a closed set of a compact space D_i^n under a continuous map into a Hausdorff space. For $n > m$ we use a variant of the closedness characterisation, which is available in mathlib as `Topology.RelCWComplex.isClosed_of_isClosed_inter_openCell_or_isClosed_inter_closedCell`. By this it suffices to consider the intersection of S with the open cells e_i^n . Since $S \subseteq X_m$ is disjoint from every open cell of dimension greater than m , these intersections are empty and hence closed. $\square$

By ?? and theorem 2.16, the map $P_m \times id_I : Z_m \times I \rightarrow X_m \times I$ is a quotient map. We now define

$$f_m : Z_m \times I \rightarrow (X_m \times \{0\}) \cup (X_{m-1} \times I)$$

$$(z, t) \mapsto \begin{cases} (P_m \times id_I) \circ r_i^m(x), & z \in D_i^m \text{ for some } i \in I_m \\ (P_m \times id_I)(z, t), & \text{otherwise} \end{cases}$$

. Here $r_i^m : D_i^m \times I \rightarrow D_i^m \times \{0\} \cup S_i^{m-1} \times I$ is the retraction, obtained from theorem 2.14 and proposition 2.7. Note that the case distinction is unambiguous: the summands of Z_m are disjoint, so z lies in a ball of dimension exactly m for at most one index i . Once we know that f_m is continuous and factors through $P_m \times id_I$, the universal property of the quotient yields a map r_m^{Skel} making the following diagram commute.

$$\begin{array}{ccc} Z_m \times [0, 1] & \xrightarrow{f_m} & X_m \times \{0\} \cup X_{m-1} \times [0, 1] \\ \downarrow P_m \times id_{[0, 1]} & \nearrow \exists r_m^{\text{Skel}} & \\ X_m \times [0, 1] & & \end{array}$$

It is not clear a priori that this map is a retraction. We verify this separately in ??.

Lemma 2.18. *For every $m \in \mathbb{N}$, the map f_m*

(i) takes values in $(X_m \times \{0\}) \cup (X_{m-1} \times I)$,

(ii) is continuous, and

(iii) factors through $P_m \times id_I$

Proof. (i). We treat the two cases of the definition separately. First let $(z, t) \in D_i^m \times I$ for some $i \in I_m$. Since r_i^m is a retraction,

$$r_i^m(z, t) \in (D_i^m \times \{0\}) \cup (S_i^{m-1} \times I).$$

If the second coordinate of $r_i^m(z, t)$ is 0, then $f_m(z, t) \in X_m \times \{0\}$. Otherwise the first coordinate lies in $S_i^{m-1} = \partial D_i^m$, and by condition (3) of definition 2.11 its image under Q_i^m lies in X_{m-1} , so $f_m(z, t) \in X_{m-1} \times I$.

Now let $(z, t) \in Z_m \times I$ be a point of the second case, so $z \in C$ or $z \in D_i^n$ for some $n < m$. In either case $P_m(z) \in X_{m-1}$, hence $f_m(z, t) \in X_{m-1} \times I$.

(ii). The space $Z_m \times I$ is homeomorphic to $(C \times I) \amalg \coprod_{n \leq m} \coprod_{i \in J_n} (D_i^n \times I)$, via the canonical mapp. Under this homeomorphism, f_m is defined on a coproduct and not a product space anymore. f_m restricts on each summand to a composite of continuous maps, and a map out of a coproduct is continuous if and only if all its restrictions to the summands are. Hence f_m is continuous.

(iii). Let (z, t) and $(z', t') \in Z_m \times I$ with $P_m \times id_I(z, t) = P_m \times id_I(z', t')$. Since $P_m \times id_I$ is the identity on the second coordinate, we directly get $t = t'$ and $P_m(z) = P_m(z')$. We must show, that $f_m(z, t) = f_m(z', t)$.

If neither z nor z' lies in a ball of dimension m , both are in the second case of the definition and $f_m(z, t) = (P_m \times id_I)(z, t) = (P_m \times id_I)(z', t) = f_m(z', t)$. The same conclusion holds if z or z' lie in ∂D_i^m for some $i \in I_m$, since r_i^m is a retraction it is the identity there and f_m again agrees with $P_m \times id_I$. So we may assume that $z \in \mathring{D}_i^m$ for some $i \in I_m$, and it suffices to show $z = z'$. We have $P_m(z) = Q_i^m(x) \in e_i^m$. Suppose z' did not lie in $\mathring{D}_i^m$. If $z' \in \mathring{D}_j^m$ for some $j \neq i$, then $P_m(z') \in e_j^m$, which is disjoint from e_i^m by condition (2) of definition 2.11. In all remaining cases — that is, if $z' \in C$, or z' lies in a ball of dimension less than m , or in the boundary of a ball of dimension m — condition (3) gives $P_m(z') \in X_{m-1}$, which is disjoint from e_i^m as well. Both contradict $P_m(z) = P_m(z')$, so $z' \in \mathring{D}_i^m$. Since $Q_i^m|_{\mathring{D}_i^m}$ is a homeomorphism onto e_i^m and hence injective, $P_m(z) = P_m(z')$ forces $z = z'$. $\square$

Theorem 2.19. *Let (X, C) be a relative CW-complex with skeleta $(X_m)_{m \in \mathbb{N}}$, then (X_m, X_{m-1}) has the HEP for every $m \in \mathbb{N}$.*

Proof. By lemma 2.18 and the universal property of the quotient we have a continuous map $r_m^{\text{Skel}}: X_m \times I \rightarrow (X_m \times \{0\}) \cup (X_{m-1} \times I)$ with $r_m^{\text{Skel}} \circ (P_m \times id_I) = f_m$. By proposition 2.7 it remains to show that r_m^{Skel} restricts to the identity on $(X_m \times \{0\}) \cup (X_{m-1} \times I)$. Then the pair then has the HEP, since X_{m-1} is closed in X_m .

Since f_m factors through $P_m \times id_I$, the value $r_m^{\text{Skel}}(x, t)$ does not depend on the choice of a preimage of (x, t) , and we may therefore choose a convenient one in each case.

Points of $X_{m-1} \times I$. Let $(x, t) \in X_{m-1} \times I$. Then x lies in C or in a closed cell of dimension less than m , so we may choose a preimage (z, t) with $z \in C$ or $z \in D_i^n$ for some $n < m$. Such a z falls under the second case of the definition of f_m . Therefor

$$r_m^{\text{Skel}}(x, t) = r_m^{\text{Skel}}((P_m \times id_I)(z, t)) = f_m(z, t) = (P_m \times id_I)(z, t) = (x, t).$$

Points of $X_m \times \{0\}$. Let $(x, 0) \in X_m \times \{0\}$. If x lies in X_{m-1} , the previous case applies. Otherwise x lies in an open cell of dimension m , so there are $i \in I_m$ and $z \in D_i^m$ with $P_m(z) = x$. Since r_i^m is a retraction and $(z, 0) \in D_i^m \times \{0\}$, we get $r_i^m(z, 0) = (z, 0)$ and therefore

$$r_m^{\text{Skel}}(x, 0) = f_m(z, 0) = (P_m \times id_I)(r_i^m(z, 0)) = (P_m \times id_I)(z, 0) = (x, 0). \quad \square$$

2.2.3 HEP for relative CW complexes

In this final section we prove that every relative CW-complex (X, C) has the HEP. For the finite dimensional complex this follows quickly from what we established so far.

Lemma 2.20. *Let (X, C) be a relative CW-complex. Then (X_n, C) has the HEP for every $n \in \mathbb{N} \cup \{-1\}$.*

Proof. We proceed by induction on n .

IBase case $n = -1$: Note, that $X_{-1} = C$, so the pair in question is $(X_{-1}, C) = (C, C)$, which has the HEP by lemma 2.2.

Inductive step $n \rightarrow n + 1$. Assume, we know that (X_n, C) has the HEP. By theorem 2.19 the pair X_{n+1}, X_n has the HEP as well, and since $C \subseteq X_n \subseteq X_{n+1}$, proposition 2.9 yields the HEP for (X_{n+1}, C) . $\square$

Corollary 2.21. *Let (X, C) be a finite dimensional relative CW complex. Then (X, C) has the HEP.*

Proof. Since (X, C) is finite dimensional, there is an $n \in \mathbb{N} \cup \{-1\}$, such that X is equal to X_n . The claim therefore follows from lemma 2.20 applied to X_n . $\square$

This is the point at which our formalisation stops. For the sake of completeness we sketch the proof of the general case, which we did not formalise in Lean.

Lemma 2.22. *Let (X, C) be a relative CW complex. Then (X, C) has the HEP.*

Sketch. We use the retraction criterion, which applies since C is closed in X . By lemma 2.20 and proposition 2.7 there are retractions

$$r_m: X_m \times I \rightarrow (X_m \times \{0\}) \cup (C \times I)$$

for all $m \in \mathbb{N} \cup \{-1\}$. These can be constructed inductively: for $m = -1$ we take the identity, and for $m \geq 0$ we let r_m be the composite

$$X_m \times I \xrightarrow{r_m^{\text{Skel}}} (X_m \times \{0\}) \cup (X_{m-1} \times I) \xrightarrow{id \cup r_{m-1}} (X_m \times \{0\}) \cup (C \times I),$$

where the second map is the identity on $X_m \times \{0\}$ and r_{m-1} on $X_{m-1} \times I$. A straightforward induction shows that these two descriptions agree on the overlap and that $r_m|_{X_{m-1} \times I} = r_{m-1}$ for all m . This compatibility allows us to define

$$r: X \times I \rightarrow (X \times \{0\}) \cup (C \times I), \quad r(x, t) := r_m(x, t) \quad \text{for any } m \text{ with } x \in X_m,$$

which is independent of the choice of m . It restricts to the identity on $(X \times \{0\}) \cup (C \times I)$, since each r_m does. Continuity follows from proposition 2.23 below, applied to $Z := I$: the restriction of r to $X_n \times I$ is r_n and hence continuous for every n . Thus r is a retraction, and the retraction criterion gives the HEP for (X, C) . $\square$

Proposition 2.23. *Let (X, C) be a relative CW-complex, let Z be a locally compact space and let Y be a topological space. A map $f: X \times Z \rightarrow Y$ is continuous if and only if the restriction $f|_{X_n \times Z}$ is continuous for every $n \geq -1$.*

We state proposition 2.23 without proof. It expresses that X carries the colimit topology of its skeleta and that this is preserved under taking the product with a locally compact space, which is the same phenomenon that theorem 2.16 describes for a single quotient map.

3 Lean and mathlib

Having discussed the informal mathematics behind our formalisation, we now give a brief overview of Lean and of the conventions and concepts that are relevant to this project.

Lean is a proof assistant, that is, a tool that checks the individual reasoning steps of a proof written in its formal language. It verifies that assumptions, axioms, functions and theorems are applied only where permitted and that new results follow by valid logical inference. According to its main designers, this verification rests upon a “small trusted kernel based on dependent type theory” (leanTP), a notion we explain in the next section. Lean is one of several proof assistants, alongside Rocq and Isabelle. As Kontorovich describes in *The Shape of Math to Come* [Kon25], it is remarkable how quickly Lean has gained relevance not only in computer science and mathematical logic, but in mainstream mathematical research.

The first version of Lean was created by Leonardo de Moura in 2013, and the first stable release of Lean 4 appeared in September 2023. Since July 2023 the development of the language has been coordinated by the non-profit *Lean Focused Research Organization*, whose stated goal is “enhanced scalability, usability, documentation, and proof automation while guiding Lean toward long-term self-sustainability” [FRO]. Alongside the language, the mathematical library *mathlib* has grown considerably: as of August 2026 it comprises almost 2.500.000 lines of code contributed by over 770 authors [1].

3.1 Lean’s Dependent Type Theory

The following description is based on the chapter *Dependent Type Theory* in the digital book *Theorem Proving in Lean 4* [Avi+24].

Every expression in Lean has a type, which can be displayed with the command `#check`. The familiar mathematical objects $\mathbb{N}$, $\mathbb{R}$, $\mathbb{C}$ and the Euclidean spaces all have type `Type`, and a function from α to β has type $\alpha \rightarrow \beta$. Types themselves are expressions and therefore have types again: `Type` has type `Type 1`, and in general `Type k` has type `Type (k+1)`. This hierarchy of *universes* is the reason why declarations in later sections carry universe variables such as `u`, introduced automatically when we write `Type*` for a type in an arbitrary universe.

```
#check  $\mathbb{N}$ 
 $\mathbb{N}$  : Type
#check EuclideanSpace  $\mathbb{R}$  (Fin 3)
EuclideanSpace  $\mathbb{R}$  (Fin 3) : Type
#check Type
Type : Type 1
#check 2 = 1
2 = 1 : Prop
```

A distinguished type is **Prop**, the type of propositions. Note that $2 = 1$ is a well-formed term of type **Prop**. Propositions need not be true.

What makes this a *dependent* type theory is that a type may depend not only on other types, but on arbitrary terms. We have already seen an instance of this: $\text{Fin} : \mathbb{N} \rightarrow \text{Type}$ assigns to every natural number n the type with n elements, so that $\text{Fin } 3$ and $\text{Fin } 5$ are distinct types determined by the values 3 and 5.

3.2 Sets and Subtypes

Type theory allows us not only to define elements of a type, but to build new types from existing ones. The most relevant construction to this project is that of a subtype. Given a type α and a predicate $p : \alpha \rightarrow \text{Prop}$, the subtype $\{x : \alpha \mid p \ x\}$ is the type whose terms are pairs consisting of a term $x : \alpha$ together with a proof of $p \ x$. Constructing such a term therefore requires supplying both components. In the example below, the zero of the closed unit ball in $\mathbb{R}$. In the example below, the zero of the closed unit ball in $\mathbb{R}$ is given by the real number 0 together with a proof that it lies in the ball. While one would not usually distinguish between this point and the real number zero, in Lean they are terms of different types.

```
#check ((0 , Metric.mem_closedBall_self zero_le_one) : Metric.closedBall
↪ (0 : ℝ) 1)
      (0, ...) : { x // x ∈ Metric.closedBall 0 1 }
```

Closely related is the type $\text{Set } \alpha$ of subset of α , which is defined as $\alpha \rightarrow \text{Prop}$. These two notions are connected by a coercion. Every $S : \text{Set } \alpha$ can be read as a type $\uparrow S$, namely the subtype $\{x : \alpha \mid x \in S\}$, and conversely every term of $\uparrow S$ has an underlying term of α , obtained by `Subtype.val`. Both directions are inserted automatically by Lean where needed.

The distinction matters for us because it offers different ways of formalising a pair of spaces (X, A) with $A \subseteq X$, two of which are used in this project. In the first, the larger space is a type X and the subspace is a term $A : \text{Set } X$. In the second, both are terms of $\text{Set } Y$ for an ambient type Y . In either case a subtype appears as soon as we want to treat a set as a space in its own right. Much of the design of the definitions below is aimed at avoiding subtypes and coercions.

3.3 Structures

A *structure* in Lean bundles several components into a single type. Its declaration lists a number of fields, each with its own type, and a term of the structure is obtained by supplying a term for every field. Structures are how mathlib organises mathematical objects: a topological space, for instance, is a structure whose fields consist of a collection of open sets together with

proofs of the axioms it has to satisfy. We use this mechanism in two places, in both cases as a structure whose fields are all propositions. Since proofs of a proposition are indistinguishable in Lean, such a structure carries no data beyond the fact that its conditions hold. It simply gives a name to a conjunction of conditions, so that a single hypothesis can be passed around instead of four separate ones, and each condition can be retrieved by its field name.

3.4 Junk Values

If a formalised statement involves a division by zero, Lean returns the result 0.

```
example : (1 : ℚ) / 0 = 0 := by norm_num
```

This makes division a total function $\mathbb{Q} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}$, even though certain inputs have no genuine mathematical meaning. As explained in [MLS25], such *junk values* are common practice in Lean. The same convention applies, for instance, to the inversion of square matrices, which is defined for singular matrices as well and returns the zero matrix, or to the derivative of a function at a point where it is not differentiable, which is defined to be zero. At first glance this may seem concerning, since one can state and prove propositions that in classical mathematics would look false or ill-defined. The issue is resolved by adding suitable hypotheses such as $n \neq 0$ or $\text{Differentiable } \mathbb{k} f$ to any statement about the object in question. These ensure that a theorem can only be applied where its conclusion is meaningful. What is gained in return is that definitions and statements become simpler, because no side conditions have to be carried along merely to make an expression well formed. We make frequent use of this convention. Functions that are mathematically defined only on a subset are defined on the entire ambient space. This allows us to extend them without subtype coercions and to apply arguments to them, without proving every time, they lie in the right subset. Junk values also appear in the formalisation of theorem 2.14: the term $x/\|x\|$ occurring in the definition of h is meaningless at $x = 0$, but since division by zero is defined to be zero, no separate treatment of this case is required.

3.5 mathlib

mathlib is a community-driven library of mathematics formalised in Lean. It contains definitions and theorems from a wide range of mathematical fields, and it is continually extended by contributors all over the world. Results are stated as generally as the setting allows, so that they are applicable in as many situations as possible. For this project, we import, among others, the topology packages, which include the formalisation of CW-complexes and some of the basic results about them that our proof relies on.

Names in `mathlib` are chosen to be self-explanatory, in the sense that the name of a result describes its statement. The theorem that a composition of two continuous functions is continuous, for example, is called `Continuous.comp`. Given `f` and `g` together with proofs `hg : Continuous g` and `hf : Continuous f`, we obtain a proof of `Continuous (g ∘ f)` by writing `Continuous.comp hg hf`, or, in the shorter and more common form using dot notation, `hg.comp hf`. This is possible, because `hg` has type `Continuous g` and the theorem's name begins with `Continuous`, so Lean resolves `hg.comp` to `Continuous.comp` with `hg` inserted as the first explicit argument of that type.

4 Lean formalisation of the Homotopy Extension Property

4.1 CW-complexes in Mathlib

1. NOTIZEN:
2. Zellen im CW complex sind würfel anstatt spheres
3. CW -complexe leben in einem größeren Raum ->

4.2 HEP Definition

Already in the mathematical part, the definition of the homotopy extension property was long and technical and its formalisation is no shorter. Throughout the formalisation project we use the following abbreviations, and for this section we fix the variables below.

```

universe u
variable {X Y : Type u} [TopologicalSpace X] [TopologicalSpace Y] {A : Set X}

abbrev cyl (X : Type*) : Set (X × ℝ) := {p | p.2 ∈ unitInterval}
abbrev smalcyl {X : Type*} (A : Set X) : Set (X × ℝ) := {p | p.1 ∈ A ∧ p.2 ∈
  ↪ unitInterval}
abbrev anchor {X : Type*} (A : Set X) : Set (X × ℝ) :=
  {p | p.2 = 0 ∨ p.1 ∈ A ∧ p.2 ∈ unitInterval}
abbrev smalanchor {Y : Type*} (X A : Set Y) : Set (Y × ℝ) :=
  {p | p.1 ∈ X ∧ p.2 = 0 ∨ p.1 ∈ A ∧ p.2 ∈ unitInterval}

```

These four abbreviations name the subsets that occur in the definition of the HEP and in the retraction criterion. The set `cyl X` is the cylinder of height one over the type `X`, whereas `smalcyl A` is the cylinder only over a set `A` of type `Set X`. The set `anchor A` is the union of the cylinder over `(A : Set X)` together with the bottom of the cylinder `cyl X`. Finally, `smalanchor X A` is the corresponding set for the case that `(X A : Set Y)` are subsets of an ambient type `Y`. It is the union of the cylinder over `A` with the bottom of `smalcyl X`. All four are declared as `abbrev` rather than `def` and are therefore reducible, so that elaboration and `simp` see through them.

We can now turn to the definition itself.

```

def agreeOn (f : X → Y) (H : X × ℝ → Y) (A : Set X) : Prop := ∀ (a : X), a ∈ A
  ↪ → f a = H (a, 0)

structure HomotopyExtension (H' : X × ℝ → Y) (f : X → Y) (H : X × ℝ → Y) (A :
  ↪ Set X) (rangeH' : Set Y) : Prop where
  ContinuousOn : ContinuousOn H' (cyl X)
  range : (cyl X).MapsTo H' rangeH'
  agreef : (∀ (x : X), f x = H' (x, 0))

```

```

agreeH : (∀ (a : A) (t : unitInterval), H (a,t) = H' (a, t))

def HEP (X : Type u) [TopologicalSpace X] (A : Set X) : Prop :=
  ∀ (Y : Type u) [TopologicalSpace Y] (rangeH' : Set Y), ∀ (f : X → Y), ∀ (H :
    ↪ X × ℝ → Y), Continuous f → Set.range f ⊆ rangeH' → ContinuousOn H
    ↪ (smalcyl A) → (smalcyl A).MapsTo H rangeH' → agreeOn f H A → ∃ H' : X ×
    ↪ ℝ → Y, HomotopyExtension H' f H A rangeH'

```

To shorten the definition and make it more readable, we first define `agreeOn`. It expresses that the map `f` and the homotopy `H` we wish to extend take the same values on `A`, respectively `A × {0}`, so that the two maps are compatible and asking for a common extension makes sense. Secondly, the structure `HomotopyExtension` bundles the properties the desired homotopy should have: continuity on the cylinder, that its values lie in the prescribed set, and that it agrees with the two given maps. Using a structure rather than a conjunction has the advantage that it comes with projections for the individual fields, which is what makes the applications in proposition 2.7 readable.

This definition of `HEP` is close to the informal definition 2.1, apart from the two aspects elaborated below. We will then present the two variants `HEPY` and `HEP'` and discuss their usefulness. Before tuning to that and as a first sanity check that the definition is usable at all, we prove that every space has the `HEP` relative to itself and relative to the empty set. The lemma `HEP_self` will serve as the base case of the induction in ??.

```

lemma HEP_self : HEP X (@Set.univ X) := by ...

```

```

lemma HEP_empty : HEP X ∅ := by ...

```

Homotopies on $X \times \mathbb{R}$

The homotopies are defined on $X \times \mathbb{R}$ rather than on $X \times \text{unitInterval}$ or even $A \times \text{unitInterval}$. The reason is that a homotopy defined on $\uparrow A \times \uparrow \text{unitInterval}$ is applied to pairs $(\langle a, ha \rangle, \langle t, ht \rangle)$: every time we want to evaluate it we have to supply a proof that the point lies in the right subset, and every equation between such terms has to be passed through `Subtype.ext` or a congruence lemma before the underlying points can be compared. On $X \times \mathbb{R}$ none of this bookkeeping appears.

This does not change the object being defined. The field `ContinuousOn H' (cyl X)` constrains `H'` only onto the cylinder. So what `H'` does outside the unit interval is irrelevant. This field shifts the focus onto exactly what the definition needs. The choice not to work with subtypes is not free of cost. In the backward direction of lemma `retraction_criterion_closed'` we will have to precompose with a projection onto $[0, 1]$ in order to make the `MapsTo` field applicable, as described in ??.

The set `rangeH'` and the variant `HEPY`

The set `rangeH'` has no equivalent in the informal definition definition 2.1. It is a set in the codomain type Y in which the extended homotopy H' is required to take its values. It forces two additional hypotheses `Set.range f  $\subseteq$  rangeH'` and `(smalcyl A).MapsTo H rangeH'`. Without them there could be no such extension, since H' extends both f and H .

The gain of carrying this set along is that it lets us apply the HEP with a *subset* of a type as codomain without ever forming the corresponding subtype. The application we have in mind is the retraction criterion: there the HEP is applied with $Y := X \times \mathbb{R}$ and `rangeH' := anchor A`, so that the extension it produces is a self-map of $X \times \mathbb{R}$ whose image lies in the anchor. This is precisely what the formalisation of a retraction onto the anchor requires, as discussed later on.

In most applications one takes `rangeH'` to be the whole codomain, written `@Set.univ Y` in Lean. The two additional hypotheses become statements that `tauto` solves. Since this extra work is negligible, we prefer this definition over the shorter variant discussed next.

Because `rangeH'` has no informal counterpart, one could ask whether it changes the notion being defined. To settle this we also formalise the literal reading of definition 2.1, in which the extension is only required to be a map into Y

```
structure HomotopyExtensionY (H' : X  $\times$   $\mathbb{R}$   $\rightarrow$  Y) (f : X  $\rightarrow$  Y) (H : X  $\times$   $\mathbb{R}$   $\rightarrow$  Y) (A :
 $\hookrightarrow$  Set X) : Prop where
  ContinuousOn : ContinuousOn H' (cyl X)
  agreef :  $\forall$  (x : X), f x = H' (x, 0)
  agreeH :  $\forall$  (a : A) (t : unitInterval), H (a,t) = H' (a, t)

def HEPY (X : Type u) [TopologicalSpace X] (A : Set X) : Prop :=
   $\forall$  (Y : Type u) [TopologicalSpace Y],  $\forall$  (f : X  $\rightarrow$  Y),  $\forall$  (H : X  $\times$   $\mathbb{R}$   $\rightarrow$  Y),
   $\hookrightarrow$  Continuous f  $\rightarrow$  ContinuousOn H (smalcyl A)  $\rightarrow$  agreeOn f H A  $\rightarrow$   $\exists$  (H' : X  $\times$ 
   $\hookrightarrow$   $\mathbb{R}$   $\rightarrow$  Y), HomotopyExtensionY H' f H A
```

The lemma `HEP_iff_HEPY` shows that the two definitions are equivalent, so that `rangeH'` is a matter of convenience and not of content. We will not use `HEPY` again in this project. It's only reason to exist is this equivalence.

Universes

In both versions of the definition the codomain Y is quantified over `Type u`, the same universes as X . This is forced by the way we use the definition.

A quantifier `$\forall$  (Y : Type v)` can only be filled with codomains lying in exactly that universe. All codomains we ever need, however, are built from the space X itself and hence live in its universe. In the retraction criterion, for instance,

the codomain is $X \times \mathbb{R}$, which lies in `Type u`. Quantifying over a lower or higher universe would make the definition inapplicable to such instances. There is a short argument why the definition with this restriction on the universe of Y is equivalent to a more general one: the retraction criterion shows that `HEP`, at least in the case that the subspace is closed, is equivalent to the existence of a retraction, and that statement does not mention the codomain Y at all.

The variant `HEP'`

Let us now turn to the variant `HEP'`, which changes the shape of the pair of spaces.

```
open Set.Notation
def HEP' {Y : Type*} [TopologicalSpace Y] (X A : Set Y) : Prop := HEP X (X ↯ A)
```

The notation $(X \downarrow A)$ means, that we regard $(X \cap A)$ as a set of type `Set ↯ X`. The difference is that in `HEP` the set A is a subset of the type X , whereas in `HEP'` both X and A are subsets of one ambient type Y .

Two consequences of defining `HEP'` through `HEP` are worth noting. First, `HEP'` does not require A to be a subset of X . To be precise, we do not check, that the pair (X, A) has `HEP`, but instead the pair $(X, X \cap A)$ does, and this is the intended statement only if $A \subset X$. In some proofs, such as `HEP_trans`, the inclusion therefore appears as an explicit hypothesis.

Secondly, `HEP` and `HEP'` are definitionally equal, so results transfer between them for free, als we will see for example when `HEP_disc_boundary'` is proved by `HEP_disc_boundary` alone.

We introduced `HEP'`, when we came to pairs of consecutive skeleta, since a skeleton of a relative CW-complex is a subcomplex whose carrier is a subset of an ambient space. So this is the shape in which the objects actually arise. It also turned out to be the better fit for balls and spheres: the sphere is then simply the set of points of norm one in the ambient normed space, rather than a subset of the subtype of the closed ball.

Note, that in this definition the maps f and h to be extended are defined on the subtype `↯ X`, respectively `↯ X × ℝ`. Working with these subtypes caused some extra work in the proof of `retraction_criterion_closed'` and offers `HEP_trans'`. One could avoid it by not defining `HEP'` in terms of `HEP` but writing it out with maps defined on Y and $Y \times \mathbb{R}$. The definition would be longer, since for instance continuity of f would become to a `ContinuousOn` statement. Also, the equivalence with `HEP` would have to be proven rather than holding by definition. Nevertheless, it is an interesting modification that should be considered looking for the best formalisation, even though we do not pursue it further in this work.

Summary

To sum up, there are the two versions HEP and HEP', both of which are used in this formalisation project. HEP is applied to a type together with a subset of it, whereas for HEP' both spaces are subsets of the same ambient type. We consider the primed version HEP' to be more usefull one to work with, optionally with the modification discussed above of spelling it out with maps defined on the ambient space. This does not mean that HEP' is preferable on counts, since the statements become longer and also the proof of the retraction criterion would have been longer and more tedious.

4.3 Retraction Criterion

Before turning to the formalisation of the retraction criterion itself, we look at the definition of a retraction. Classically, a retraction r from B onto A is defined via precomposition with the inclusion $A \hookrightarrow B$, as in definition 2.5. As with the definition of the HEP, we again want to avoid working with subtypes. Instead of defining a map from B to A , we therefore let the retraction be a map from an ambient type X to itself. The structure `RetractionOn` bundles the properties that make such a map a retraction.

```
structure RetractionOn (r : X → X) (B A : Set X) : Prop where
  subset : A ⊆ B
  continuousOn : ContinuousOn r B
  mapsTo : B.MapsTo r A
  fixesOn : ∀ a ∈ A, r a = a
```

As the name `RetractionOn` suggests, the map r is a self-map of the type X that is not a retraction on all of its domain. It only "retracts" a subset B onto a subset A . The four fields translate the classical conditions as follows. The first property `subset` states that A is contained in B . Note, that omitting this property would define a more general class of self-maps that need not staisfy $r(B) = A$. The field `continuousOn` requires continuity on the relevant subset B only. Since the codomain of a retraction is supposed to be A , the field `mapsTo` requires that r sends every element of B to an element of A . Finally, the classical definition demands that precomposition with the inclusion $A \hookrightarrow B$ gives the identity id_A . In our setting there is no inclusion map, since domain and codomain of r are the same type X , so the condition therefore becomes the requirement that ever element of A is fixed by r . This is the field `fixesOn`.

Recall that the retraction criterion was obtained from two lemmas, and the formalisation follows the same structure. Since lemma 2.4 consists of two implications, only one of which requires closedness, we split it into two separate lemmas, `if_HEP_then_extension` and `retraction_criterion_closed` which

is the equivalence under closedness assumption, so that the closedness assumption appears only where it is needed. Their proofs are unspectacular and follow the informal argument of lemma 2.4 closely, as do the formalisations of lemma 2.6 and of the retraction criterion proposition 2.7 itself.

```
lemma retraction_criterion_closed (hA1 : IsClosed A) : HEP X A ↔
  ∃ r : X × ℝ → X × ℝ, RetractionOn r (cyl X) (anchor A) := by ...
```

Here the retraction is a map on the product $X \times \mathbb{R}$, where X is the larger space of the pair. Although the lemma is phrased in terms of HEP, it can be applied to HEP' as well, since the two definitions are definitionally equal:

```
variable (X A : Set Y) (hHEP' : HEP' X A)
#check if_HEP_then_retraction hHEP'
if_HEP_then_retraction hHEP' : ∃ (r : ↑X × ℝ → ↑X × ℝ), RetractionOn r (cyl ↑X)
  ↪ (anchor (X ↓∩ A))
```

While this does yield a retraction for the spaces in question, it is not formulated in the spirit of HEP': the retraction still lives on $X \times \mathbb{R}$ for a type X , whereas HEP' is stated for subsets of an ambient type. We therefore also formalised the lemma `retraction_criterion_closed'`, in which the retraction is a map from the ambient space $\gamma \times \mathbb{R}$ to itself. As before, the Lean file contains the forward implication without the closedness assumption.

```
lemma retraction_criterion_closed' (A X : Set Y) (hAX : A ⊆ X)
  (hA1 : IsClosed A) : HEP' X A ↔ ∃ s : Y × ℝ → Y × ℝ, RetractionOn s
  ↪ (smalcyl X) (smalanchor X A)
:= by ...
```

At this point the ambient-space formulation introduces a discrepancy with the informal statement, which went unnoticed during the formalisation. In lemma 2.4 and proposition 2.7 the hypothesis is that A is closed in X , that is, in the subspace topology of X . In `retraction_criterion_closed'`, however, both A and X are subsets of the ambient type γ , and `hA1 : IsClosed A` asserts that A is closed in γ . This is strictly stronger. In practice this turned out to be convenient rather than restrictive, since all closedness facts we need are available in the ambient space anyway: the skeleta of a relative CW-complex are closed in the ambient type, as is the sphere S^{m-1} in $\mathbb{R}^m$. Ideally one would drop the assumption entirely. Hatcher [Hat02] states the retraction criterion without it, the general case being proved in the appendix following an argument of [Str68]. Formalising that argument would remove the discrepancy, but it is considerably more involved than the proof we give.

We proved `retraction_criterion_closed'` from `retraction_criterion_closed`. For the forward direction we obtain a retraction r as in the `#check`

above. In order to extend this map $r : \uparrow X \times \mathbb{R} \rightarrow \uparrow X \times \mathbb{R}$ to a map $s : Y \times \mathbb{R} \rightarrow Y \times \mathbb{R}$, we precomposed in the first component with a map $\text{mapYX} : Y \rightarrow X$, which is the identity on X and sends all other points of Y to an arbitrary junk value, and postcomposed with Subtype.val , which sends $(r\ x).1 \in X$ back to the corresponding point of Y . It remains to confirm that this newly defined map is again a retraction, that is, that it is continuous on the relevant set, maps into the required one, and leaves the necessary values unchanged.

For the backward direction we start with a retraction $s : Y \times \mathbb{R} \rightarrow Y \times \mathbb{R}$ and want to construct $r : \uparrow X \times \mathbb{R} \rightarrow \uparrow X \times \mathbb{R}$, so that `retraction_criterion_closed` gives the claim. One might expect this direction to be the easier one, since no new values have to be constructed and s only needs to be restricted to the desired space. This expectation is disappointed at the point where the codomain of r is required to be $X \times \mathbb{R} \subseteq Y \times \mathbb{R}$. All we know about the range of s is that its codomain is $Y \times \mathbb{R}$ and that it is a retraction on `smalcyt` X , so the field `RetractionOn.mapsTo` gives no information about where in $Y \times \mathbb{R}$ a point is sent once its second coordinate lies outside the unit interval. We resolved this by precomposing the real coordinate with `ProjIcc` $: \mathbb{R} \rightarrow \mathbb{R} := \text{fun } t \mapsto \text{Set.projIcc } (0 : \mathbb{R})\ 1\ \text{zero_le_one } t$, where `Set.projIcc` is a continuous map from `mathlib` that retracts a linearly ordered type onto a closed interval in it. After this precomposition the `mapsTo` field of s is applicable to every point we feed into it. This extra work is a consequence of working with types rather than subtypes; it could have been avoided by defining homotopies on $Y \times \text{unitInterval}$ instead of $Y \times \mathbb{R}$.

The two versions of the criterion differ in two independent aspects, which are worth keeping apart. The first is which definition of HEP appears in the statement. As `#check` above shows, this is a matter of style. `HEP` and `HEP'` are definitionally equal, so either criterion could have been stated in terms of either definition. The second aspect is where the retraction lives, on $\uparrow X \times \mathbb{R}$ for a type X or on $Y \times \mathbb{R}$ for an ambient type Y , and here the two forms are not interchangeable. While we preferred working with types instead of subtypes for the definition of the homotopy extension property, here we experienced both versions valuable in practice. I will give an example for each version:

The unprimed version `retraction_criterion_closed` is what the proof for two consecutive skeletons needs, where we construct a retraction $r : (\text{skeletonLT } X\ m) \times \mathbb{R} \rightarrow (\text{skeletonLT } X\ m) \times \mathbb{R}$. The key ingredient for the continuity of this map is `Topology.IsQuotientMap.continuous_lift_prod_left`, `mathlib`'s formalisation of theorem 2.16 on products of quotient maps with a locally compact factor. Working with an ambient space here would have destroyed the surjectivity of the quotient map, and the theorem could not have been applied in the form in which `mathlib` provides it.

The primed version on the other hand is what `PartialHomeomorph.HEP'` needs, which we discuss in section 4.4: a partial homeomorphism is a map between two ambient types, so a retraction defined on the ambient product can be

used more directly.

4.4 Preservation of the HEP

Besides the retraction criterion we formalised two more results that are useful for proving that a pair of topological spaces has the HEP. The first is that the HEP is preserved under partial homeomorphisms.

```
Lemma PartialHomeomorph_HEP' {X1 X2 : Set X} (hX : X2 ⊆ X1) {Y1 Y2 : Set Y}
  ↪ (hY : Y2 ⊆ Y1) (hHEP : HEP' X1 X2) (f : PartialHomeomorph X Y) (source :
  ↪ X1 = f.source) (target : Y1 = f.target) (h2 : f '' X2 = Y2) (hY2closed :
  ↪ IsClosed Y2) : HEP' Y1 Y2 := by ...
```

A `PartialHomeomorph X Y` in `mathlib` is a map between the ambient types `X` and `Y` together with a distinguished source and target on which it is a homeomorphism. The hypotheses `source`, `target` and `h2` identify the two pairs of spaces with these data, and they are stated as equations so that they can be removed by `subst` in the first step of the proof. Afterwards the whole argument is phrased in terms of `f.source`, `f.target` and `f '' X2`, and no transport along the equations is needed anywhere else. Here the ambient formulation of the retraction criterion pays off. We apply `if_HEP_then_retraction'` to `hHEP` and obtain a retraction of `smalcyl f.source` onto `smalanchor f.source X2`. Since this retraction is a self map of $X \times \mathbb{R}$, we can conjugate it with the partial homeomorphism, which is likewise a between the ambient types. In the first coordinate we precompose with `f.symm` and postcompose with `f`. If only the unprimed criterion were available, the retraction would live on $\uparrow X1 \times \mathbb{R}$, and we would first have to turn `f` into a homeomorphism $\uparrow f.source \simeq_t \uparrow f.target$ and work with subtype coercions. Note, that only `Y2` is required to be closed. This is because extracting the retraction from `hHEP` is the forward implication of the retraction criterion, which does not require closedness.

The second result is a transitivity property: if the pairs (A_1, A_2) and (A_2, A_3) both satisfy the HEP, then so does the pair (A_1, A_3) .

```
Lemma HEP_trans (A1 A2 A3 : Set X) (h21_sub : A2 ⊆ A1) (h12_hep : HEP' A1 A2)
  ↪ (h32_sub : A3 ⊆ A2) (h23_hep : HEP' A2 A3) : HEP' A1 A3 := by ...
```

For the formalisation of this proof we had a choice between two approaches. Instead of applying the retraction criterion to all three HEP statements and constructing the desired retraction from the two given ones, we proved the lemma directly from the definition. This turned out to be the shorter way, and a further advantage of the choice is that no closedness assumptions are needed, whereas applying the retraction criterion would have required them.

In retrospect, `PartialHomeomorph_HEP'` could be proven in the same manner, by checking the definition and construction the extenden homotopy explicitly rather than going through the retraction criterion. In that case the closedness assumption on `Y2` could be dropped as well. With the current definition of `HEP'`, however, in which the maps we wish to extend live on subtypes, such a proof would be less pleasant to carry out.

4.5 HEP for discs and cubes

The goal of the file `HEP_ball_cube.lean` is to prove the homotopy extension property for balls and cubes relative to their boundary. These are not only the first interesting pairs of spaces for which we want to prove the HEP, but also the first step in the proof of the *homotopy extension property* for relative CW complexes.

We start with the balls. As in the proof of `??`, we first construct the auxiliary function `aux_fun`, which combines the information of the given maps `f` and `H`. This function is then used to define the desired homotopy `H'` explicitly, with which we cab check the definition of HEP.

```
variable {m : ℕ} {Y : Type} (C : Set Y)
(f : (Metric.closedBall (0 : EuclideanSpace ℝ (Fin m)) 1) → Y)
(H : (Metric.closedBall (0 : EuclideanSpace ℝ (Fin m)) 1) × ℝ → Y)

def aux_fun : EuclideanSpace ℝ (Fin m) → Y := fun p =>
  if hp : norm p ≤ 1 then f ⟨p, by simp[hp]⟩
  else H (⟨((norm p)-1 : ℝ) • p, inv_norm_smul_mem_unitClosedBall p⟩, norm p -
    ↪ 1)

def H' : (Metric.closedBall (0 : EuclideanSpace ℝ (Fin m)) 1) × ℝ → Y :=
  fun p => (aux_fun f H) ((1 + p.2) • p.1)
```

We split the `ContinuousOn` proof of `H'` into three steps. We start with a lemma stating that `aux_fun` is continuous on the annulus $\{x \mid 1 \leq \|x\| \leq 2\} \subset \mathbb{R}^m$ and used it to prove continuity of `aux_fun` on the ball of radius two. That ball can be written as the union of the ball of radius one with the annulus, and continuity on both of these closed subsets yields continuity on their union. The continuity of `H'` then follows by composing two `ContinuousOn` functions. The proofs of the remaining properties of `H'` are not hard. Solving, rewriting and confirming inequalities makes up a large part of them. All these results combined prove the lemma `HEP_disc_boundary`. The primed version `HEP_disc_boundary'` is definitionally equal.

```
lemma HEP_disc_boundary : ∀ (m : ℕ),
  HEP (Metric.closedBall (0 : EuclideanSpace ℝ (Fin m)) 1) (Metric.sphere ⟨0,
    ↪ by simp⟩ 1) := by ...

lemma HEP_disc_boundary' : ∀ (m : ℕ),
```

```

HEP' (closedBall (0 : EuclideanSpace ℝ (Fin m)) 1) (sphere 0 1) :=
↪ HEP_disc_boundary

```

The `Metric.sphere` in Lean has type `Set α`, where α is the type of the center, in our case the zero of an m -dimensional Euclidian space. By the way HEP is defined, the sphere in `HEP_disc_boundary` is a subset of the closed ball. so we have to add the `by simp` proof in order to construct the element of the subtype. For `HEP'`, on the other hand, both spaces live in the same ambient space. `Np` proof has to be supplied, and Lean interprets the zero as the origin of $\mathbb{R}^m$. Here we see an advantage of the `HEP'` definition in action, since we can reduce the use of subtypes.

In the formalisation of (relative) CW-complexes, each cell is the image of a closed ball in $\text{Fin } m \rightarrow \mathbb{R}$ for some m under a characteristic map. Like `EuclideanSpace ℝ (Fin m)`, the type $\text{Fin } m \rightarrow \mathbb{R}$ is a way of formalising an m -dimensional real vector space. The relevant difference between them is that `EuclideanSpace` comes with the Euclidean norm, whereas on the other version the default norm is the maximum norm. So the closed ball that is the preimage of a closed cell in a CW-complex is something we would usually call a cube rather than a ball. To prove the HEP for these cubes, we want to use the fact that an m -dimensional ball is homeomorphic to a cube of the same dimension.

Instead of constructiong this homeomorphism explicitly, we obtain it from the following `mathlib` lemma. It states that for a convex bounded set with nonempty interior in a real normed space there is a homeomorphism of the ambient space with itself that carries the interior, the closure and the frontier of the set onto the open unit ball, the closure and the unit sphere respectively. `ToDo indirektes Zitat aus Mathlib (auch das folgende lemma) – Quelle hinzufuegen`

```

exists_homeomorph_image_interior_closure_frontier_eq_unitBall.{u_1} {E : Type
↪ u_1} [NormedAddCommGroup E][NormedSpace ℝ E] {s : Set E} (hc : Convex ℝ s)
↪ (hne : (interior s).Nonempty) (hb : Bornology.IsBounded s) :
  ∃ h, ↑h '' interior s = ball 0 1 ∧ ↑h '' closure s = closedBall 0 1 ∧ ↑h ''
↪ frontier s = sphere 0 1

```

The `Metric.closedBall` for which we already proved the HEP, is convex, bounded and has nonempty interior, so it satisfies all three assumptions. Applying the lemma to it directly would nevertheless gain us nothing: inside `EuclideanSpace ℝ (Fin m)` this set already is the closed unit ball, so the lemma would be only produce a homeomorphism carrying it onto itself, and no cube would appear anywhere. What makes the cube appear is that in $\text{Fin } m \rightarrow \mathbb{R}$ it is the closed unit ball which is the cube. We therefore first transoort our closed ball into that space along the homeomorphism

```

def homeo_euclid_max : EuclideanSpace ℝ (Fin m) ≈ₜ (Fin m → ℝ) where
  toFun := fun p ↦ p
  ...

```

The map `homeo_euclid_max` sends each point to itself and only changes the norm of the space. The image of the `Metric.closedBall` is therefore still the same round convex set, even though it is no longer the closed unit ball of the codomain. Applying the lemma to this image yields a homeomorphism of $\text{Fin } m \rightarrow \mathbb{R}$ to itself that carries the round set onto the cube:

```
def G : (Fin m → ℝ) ≈ₜ (Fin m → ℝ) :=
  ↪ (exists_homeomorph_image_interior_closure_frontier_eq_unitBall
    ↪ convex_euclid_to_max_ball nonempty_euclid_to_max_ball
    ↪ bounded_euclid_to_max_ball).choose
```

Since the lemma only asserts that such a homeomorphism exists, we use `.choose` to fix and name one of them, and its three defining properties are then available through `.choose_spec`.

The composition of these two homeomorphisms is in particular a partial homeomorphism from `EuclideanSpace ℝ (Fin m)` to $\text{Fin } m \rightarrow \mathbb{R}$ with source the closed ball and target the closed cube. Applying the lemma `PartialHomeomorph_HEP'` now proves

```
lemma HEP_cube_boundary' (m : ℕ) : HEP' (closedBall (0 : (Fin m → ℝ)) 1)
  ↪ (sphere 0 1) := by ...
```

```
lemma HEP_cube_boundary (m : ℕ) : HEP (closedBall (0 : (Fin m → ℝ)) 1) (sphere
  ↪ ⟨0, by simp⟩ 1) := by
  exact HEP_cube_boundary' m
```

Before concluding this section, we want to point out that here the definition of `HEP'` matches the structure of a partial homeomorphism particularly well. When we first proved this result for `HEP`, before `HEP'` was defined we had to show that the `Subtype.val` image of a sphere in the closed ball is the same as the sphere in the ambient space, and likewise for the closed ball itself. With `HEP'` these subtype coercions are avoided and the proof is more straightforward. From this result the statement for `HEP` follows directly.

4.6 HEP for consecutive skeletons

Now we continue with the second step in the proof of the HEP for relative CW-complexes, namely that every pair of two consecutive skeletons satisfies the HEP. The formalisation of this result builds on the following mathlib lemma, which takes the place of the theorem stating that the product of a quotient map with the identity on a locally compact space, is again a quotient map. – `ToDo` cref zum Mathe Teil

```
theorem Topology.IsQuotientMap.continuous_lift_prod_left {X₀ : Type u_1} {X :
  ↪ Type u_2} {Y : Type u_3} {Z : Type u_4} [TopologicalSpace X₀]
  ↪ [TopologicalSpace X] [TopologicalSpace Y] [TopologicalSpace Z]
  ↪ [LocallyCompactSpace Y] {f : X₀ → X} (hf : IsQuotientMap f) {g : X × Y →
  ↪ Z} (hg : Continuous fun (p : X₀ × Y) => g (f p.1, p.2)) :
  Continuous g
```

The theorem tells us that a map g defined on the product of a quotient space with a locally compact space is continuous, provided that its composition with the quotient map in the first factor is. We want this g to be the map that yields the HEP for two consecutive skeletons. Mathematically it is the retraction $X_m \times [0, 1]$ onto $X_m \times \{0\} \cup X_{m-1} \times [0, 1]$. In the formalisation it takes the shape `skeletonLT X (m+1) × unitInterval → skeletonLT X (m+1) × ℝ`, which will be discussed later on. For now, let us take a closer look at which functions we need to define and which results we have to prove in order to get there.

A remark on the indices before we start. Throughout the project we work with `skeletonLT`, which collects the cells of dimension strictly less than its argument, so that `skeletonLT X (m+1)` is the m -skeleton X_m . The parameter of the maps `r`, `sumMap` and `cubesretract`, which will be introduced in this chapter, is therefore the top dimension m of the pair, and `Fin (m+1)` is exactly the index type of the dimensions occurring in X_m . Using `skeletonLT` rather than `skeleton` also avoids truncated subtractions on $\mathbb{N}$ and converts the (-1) -skeleton, which makes it convenient for the induction in ??.

4.6.1 Skeleton is a quotient space

First of all we need a quotient map onto the skeleton `skeletonLT X m` called `f` in the theorem above. For this we use the fact that we can "unglue" the skeleton into the base space and the disjoint union of closed balls, or cubes to be precise, in all dimensions less than m . The identity map on the base space and the characteristic maps on the closed balls assemble into the map `SkeletonProjection`.

```
variable {Y : Type*} [TopologicalSpace Y] [T2Space Y] (X : Set Y) {C : Set Y}
↪ [RelCWComplex X C]

abbrev ungluedSkel (m : ℕ) :=
  C ⊕ (Σ (n : Fin m) (c : cell X n), (closedBall (0 : Fin n → ℝ) 1))

def SkeletonProjection (m : ℕ) : ungluedSkel X C m → skeletonLT X m :=
  ↪ Sum.elim (fun c ↦ ⟨c, by ...⟩) (fun ⟨n, i, x⟩ ⇒ ⟨map n i x, by ...⟩)
```

To prove that this indeed defines a quotient map, we have to check that it is surjective and continuous and that it induces the topology on the skeleton. The skeleton is a subcomplex of the relative CW-complex, and its topology is described in terms of the closed sets: A subset of the subcomplex is closed precisely when its intersection with the base space is closed there and its intersection with each closed cell is closed in that cell.

The surjectivity proof is straightforward. A point of the skeleton lies either in the base space or in an open cell, and in the two cases it has a preimage in `C` respectively in the closed ball belonging to that cell.

Continuity is not much work either, since `SkeletonProjection` is assembled from continuous functions on a disjoint union. Proving that the topology on the skeleton is the induced one required more effort.

We start with a set `S` of Type `Set (skeletonLT X ↑m)` and the assumption that its preimage under `SkeletonProjection` is closed in the unglued skeleton. The goal is to show that `S` is closed in the skeleton. In `mathlib` the results about closed sets are always phrased for subsets of the ambient space, together with the assumption that the set is contained in the CW complex or in the subcomplex. To get into this setting, we need the following lines in the proof:

```
set S' : Set Y := Subtype.val '' S with hS' def
have hS'sub : S' ⊆ (skeletonLT X m) := by
  rintro _ ⟨a, -, rfl⟩;
  exact a.2
```

Now we can apply the lemma `Topology.RelCWComplex.isClosed_of_isClosed_inter_openCell_or_isClosed_inter_closedCell`, which states that `S'` is closed in the relative CW-complex if its intersection with the base space is closed and if for every cell either `S' ∩ openCell n j` or `S' ∩ closedCell n j` is closed. We prove the case of the base space in `hClosedBase` and the case of cells of dimension less than m in `hClosedLT`. For higher dimensions the intersection of `S'` with the open cell is empty and hence closed.

4.6.2 Construction of the retraction

The next preparation is the construction of the map r , which is to be the retraction for two consecutive skeletons. As in the mathematical proof, we do not define the retraction map r on `skeletonLT X (m + 1) × [0, 1]` directly. Instead we say what it should do on the unglued skeleton, where the base space and the individual cubes are still separate, and then push the resulting map down along the quotient map `SkeletonProjection`. We build this up from the bottom: first the map on all single cubes, called `cubesretract`, then the map on the product of the whole unglued skeleton and the unit interval, which is called `sumMap` and was called f in the mathematical section, and finally r itself.

On a cube of top dimension the retraction we need is already available. It is the map `r_cube` that the retraction criterion provides from `HEP_cube_boundary`, the HEP of a cube relative to its boundary proved in `??`. On cubes of lower dimension nothing should happen. The map `cubesretract` combines the two cases. Apart from the case distinction it only coerces the second coordinate from the unit interval to $\mathbb{R}$.

```
def cubesretract {m : ℕ} (n : Fin (m + 1)) :
  (closedBall (0 : Fin (n : ℕ) → ℝ) 1) × unitInterval →
```

```

(closedBall (0 : Fin (n : ℕ) → ℝ) 1) × ℝ := fun p ↦
if n = m then (r_cube n (p.1, Subtype.val p.2))
else (p.1, Subtype.val p.2)

```

For `cubesretract` we proved lemmas about its continuity, about where it is fixed, and about the set in which its image lies. The last one comes in a version for the lower dimensions and one for the top dimension, matching the two branches of the definition.

```

lemma cubesretract_continuous {m : ℕ} (n : Fin (m + 1)) : Continuous
↔ (cubesretract n) := by ...

```

```

lemma cubesretract_fixedOn {m : ℕ} {n : Fin (m + 1)} (x : (closedBall (0 : Fin
↔ n → ℝ) 1) × unitInterval) :
(n < m → cubesretract n x = (x.1, x.2.1)) ∧
(n = m → x ∈ {z | z.2 = 0 ∨ z.1 ∈ sphere ⟨0, by simp⟩ 1} → cubesretract n
↔ x = (x.1, x.2.1)) := by ...

```

```

lemma cubesretract_mapsto_lt {m : ℕ} (n : Fin (m + 1)) (i : cell X n) (p :
↔ (closedBall 0 1) × ↑unitInterval) (hn : n < m) :
(map n i) (cubesretract n p).1 ∈ skeletonLT X m ∧ (cubesretract n p).2 ∈
↔ unitInterval
:= by ...

```

```

lemma cubesretract_mapsto_top {m : ℕ} (n : Fin (m + 1)) (i : cell X n) (p :
↔ (closedBall 0 1) × ↑unitInterval) (hn : n = m) :
(cubesretract n p).2 = 0 ∨ (map n i) (cubesretract n p).1 ∈ skeletonLT X m
↔ ∧ (cubesretract n p).2 ∈ unitInterval := by ...

```

Next we assemble these maps into a map on the product of the whole unglued skeleton times the unit interval.

```

def sumMap (m : ℕ) :
(ungluedSkel X C (m + 1)) × unitInterval → skeletonLT X (m+1) × ℝ := fun p
↔ ↦ p.1.elim
(fun c => ((c.val, memBase_memSkeletonLT X c.prop (m + 1)), p.2))
(fun ⟨n, i, x⟩ => ((map n i (cubesretract n (x,p.2))).1,
↔ cubesretract_mapsto_weak n i (x,p.2), (cubesretract n (x,p.2)).2))

```

The map `sumMap` is defined using `Sum.elim`, which performs a case analysis on a sum type and applies the corresponding function depending on which constructor is present (ToDo : indirektes Zitat. Quelle - Dokumentation von Lean), that is, on whether the first coordinate of the point `p` lies in the base space `C` or in a closed ball. In the first case we want to do nothing except apply `SkeletonProjection` to that coordinate, which on the base space is just the identity. We send the point `c : C` to itself, but regarded as an element of the skeleton. Since the proof that a point of the base space lies in any given skeleton is needed repeatedly, we made it a lemma of its own, called `memBase_memSkeletonLT`.

In the second case, where the first coordinate of p comes from a closed ball, we first apply `cubesretract` to the point and then postcompose with what `SkeletonProjection × id` would do to a such point. Since the first coordinate, still lies in a closed ball after `cubesretract` has been applied, `SkeletonProjection` amounts to applying the characteristic map of that closed ball. We have to supply a poof, that the image of this characteristic map really lies in the skeleton. This proof is again a lemma of its own, called `cubesretract_mapsto_weak`, which is derived from the two `mapsto` statements above.

It remains to push `sumMap` down from the unglued skeleton to the skeleton itself, which is what `Function.extend` does.

```
def r (m : ℕ) : skeletonLT X (m + 1) × unitInterval → skeletonLT X (m + 1) × ℝ
↪ :=
  Function.extend (fun p ↦ ((SkeletonProjection X (m + 1) p.1), p.2)) (sumMap
    ↪ X m) (Prod.map id Subtype.val)
```

`Function.extend` works in such a way that whenever a point $x \in \text{skeletonLT } X \ (m + 1)$ has a preimage $p \in \text{ungluedSkel} \times [0, 1]$ under the map $(\text{fun } p \mapsto ((\text{SkeletonProjection } X \ (m + 1) \ p.1), p.2))$, then $r \ x = \text{sumMap } X \ m$, and otherwise $r \ x = (\text{Prod.map id Subtype.val}) \ x$. This is not captured in the definition, but we know that `SkeletonProjection × id` is surjective, so the second case of `Function.extend` could be filled with any junk function, since it will never be applied. One can therefore read the definition of r simply as: take any preimage under `SkeletonProjection` and apply `sumMap` to it. That this is well defined is the content of the lemma `factors` in the next section.

It might be irritating, that r , which is supposed to be a retraction, is formalised with the unit interval as the second factor of its domain, where we would usually take the real numbers, since our retractions are defined on a product with $\mathbb{R}$. But observe that the theorem `Topology.IsQuotientMap.continuous_lift_prod_left` yields `Continuous` and not `ContinuousOn`. We should therefore not make the domain any bigger than what is mathematically relevant. In the codomain we do not have this kind of problem and can simply choose Z to be the skeleton times the entire real line.

Taking a closer look at the definition of `sumMap`, one sees that the case where p comes from the base space or from a closed ball of dimension less than m , it does the same as `SkeletonProjection × id`. At least if one ignores the subtype issues. It would have been inconvenient to write this application into the definition of `sumMap`, so instead we proved the two lemmas that state exactly this equality.

```
lemma sumMap_eq_SkelProj_C (m : ℕ) (x : C) (t : unitInterval) : sumMap m
  ↪ (Sum.inl x, t) =
```

```
Prod.map id (Subtype.val) (SkeletonProjection X (m + 1) (Sum.inl x), t) :=
  ↪ by ...
```

```
lemma sumMap_eq_SkelProj_sphereLT {m : ℕ}
  (n : Fin (m + 1)) (i : cell X n) (x' : closedBall (0 : Fin n → ℝ) 1) (t :
  ↪ unitInterval) (hx : n < m ∨ n = m ∧ x' ∈ sphere ⟨0, by simp⟩ 1) :
  sumMap m (Sum.inr ⟨n, ⟨i, x'⟩⟩, t) = Prod.map (SkeletonProjection X (m +
  ↪ 1)) Subtype.val (Sum.inr ⟨n, ⟨i, x'⟩⟩, t) := by ...
```

4.6.3 Continuity of the retraction

Now that all relevant maps are defined, we can turn to the proof that r is continuous. As announced, the continuity comes from the theorem `Topology.IsQuotientMap.continuous_lift_prod_left`. Let us see what is left to prove:

```
lemma continuousRetract (m : ℕ) : Continuous (r (X := X) m) := by
  refine (SkeletonProjectionIsQuotientMap (m + 1) X).continuous_lift_prod_left ?_
  ...
```

After this first line of the proof the current goal becomes

```
Continuous fun (p : (ungluedSkel X C m) × [0,1]) ⇒ r (SkeletonProjection
  ↪ p.1, p.2)
```

Showing this comes down to two major steps. The first is that `sumMap` factors through `(fun p ↪ ((SkeletonProjection X (m + 1) p.1), p.2))`, from which it follows easily that the continuity goal is equivalent to the continuity of `sumMap` is continuous. That continuity is then the second step.

We start with

```
lemma factors (m : ℕ) : (sumMap m).FactorsThrough (fun p ↪
  ↪ ((SkeletonProjection X (m + 1) p.1), p.2)) := by
  intro x y hxy
  ...
```

Here x and y are two points of `ungluedSkel × [0,1]` whose second coordinates agree and which have the same image under `SkeletonProjection`. We have to show that `sumMap X m x = sumMap X m y`. In `sumMap_eq_SkelProj_C`, and `sumMap_eq_SkelProj_sphereLT` we already have useful for this this goal. One part that required some work is the following:

```
lemma SkeletonProj_inj_top {m : ℕ} {n : Fin (m + 1)} (hnm : ↑n = m) (i : cell
  ↪ X n)
  (x' : closedBall (0 : Fin n → ℝ) 1) (x_ball : x'.1 ∈ ball (0 : Fin n → ℝ) 1)
  (y : ungluedSkel X (m + 1))
  (hskel : SkeletonProjection X (m + 1) (Sum.inr ⟨n, ⟨i, x'⟩⟩) =
  ↪ SkeletonProjection X (m + 1) y) :
  y = Sum.inr ⟨n, i, x'⟩ := by ...
```

The lemma `SkeletonProj_inj_top` says that if x lies in a specific open ball of top dimension and y is any other point of `ungluedSkel` with the same image under `SkeletonProjection`, then y is that very point x .

The proof of `factors` is long because it splits into many case distinctions, beginning with whether x lies in the base space or in a closed ball, and considering in each case the same two options for y separately. In the case where x lies in a closed ball we distinguish further whether it lies in the top-dimensional open cell or not. Most of these cases are quite short in themselves.

In the mathematical argument it was enough to distinguish whether a point lies in the top-dimensional cell or not. So one may ask why the formalisation splits the situation up so much further. The reason is that we cannot simply say that a point lies in the base space. A point of `ungluedSkel` is given by a constructor, either as `Sum.inl c` for some $c : C$ or as `Sum.inr ⟨n, i, x⟩`, so the case analysis over the constructors has to be made anyway, and it is then easier to treat the resulting combinations separately. – `ToDo` / `Question` : Ist das verstaendlich?

For the continuity of `SumMap` we precompose the map with the homeomorphism

```
def homeoProd (m : ℕ) :
  (C × unitInterval) ⊕ (Σ (n : Fin (m + 1)) (λ _ : cell X n),
    (closedBall (0 : Fin n → ℝ) 1) × unitInterval) ≃ₜ
  (C ⊕ (Σ (n : Fin (m + 1)) (λ _ : cell X n), closedBall (0 : Fin n → ℝ) 1)) ×
    unitInterval := ...
```

The domain of this composite is no longer a product, but a disjoint union, and on each summand the continuity goal is proved by combining results established earlier such as `cubesretract_continuous` and the continuity of the characteristic maps. Since continuity is preserved under precomposition with a homeomorphism, this yields that `sumMap`, and finally also `r`, is continuous.

4.6.4 Properties of the retraction

In this section we conclude the proof of the HEP for consecutive skeletons. For the proof via the retraction criterion we have to make a small adjustment to the map `r`. As mentioned above the desired retraction should have `(skeletonLT X (m + 1)) × ℝ` as its domain, whereas `r` is only defined on the unit interval in the second component. The easiest way to extend `r` turned out to be precomposing the second coordinate with the map `ProjIcc`, which was defined in the very first file. The final definition therefore looks as follows.

```
def retr_skeleton (m : ℕ) : (skeletonLT X (m + 1)) × ℝ → (skeletonLT X (m +
  1)) × ℝ := fun p ↦
  (r m) (p.1, ⟨ProjIcc p.2, proj_mem p.2⟩)
```

It only remains to show that `retr_skeleton` is continuous, maps into the right set and is fixed where required. The continuity follows directly from

all the work done before and the claims about `mapsTo` and `fixedOn` are first proved for `sumMap` and then transferred to `retr_skel`. (ToDo: man könnte hier noch auf `subst` eingehen, weil das ein game changer war.) Altogether this results in the following lemma.

```
lemma HEP'_skeleton (m : ℕ) : HEP' (skeletonLT X (m + 1)).carrier (skeletonLT
↪ X m).carrier := by
  apply (retraction_criterion_closed (skeletonLT X ↑m).closed'.preimage_val).2
  use retr_skeleton X m
  ...
```

Note that although the statement is phrased in terms of `HEP'` since both skeleta live in the ambient type, the proof applies the unprimed `retraction_criterion_closed`, which is possible because the two definitions are definitionally equal.

4.7 HEP for finite dimensional relative CW complexes

In this last part of the formalisation we arrive at the result that finite dimensional relative CW complexes satisfy the HEP. As the mathematical chapter already showed, the finite dimensional case follows from what we have proven so far by a straightforward induction, whereas the general case requires considerably more work. Due to the limited time for this thesis I had to stop at this point.

We first run the induction over the skeleta, and then show that a finite dimensional complex is one of its own skeleta. The induction looks in Lean as follows.

```
variable {Y : Type*} [TopologicalSpace Y] [T2Space Y] (X : Set Y) {C : Set Y}
↪ [RelCWComplex X C]

lemma HEP_skelLT_base (m : ℕ) : HEP' (skeletonLT X m) C := by
  induction m with
  | zero =>
    rw [ENat.coe_zero, skeletonLT_zero_eq_base, HEP', Subtype.coe_preimage_self]
    exact HEP_self
  | succ n n_ih =>
    apply HEP_trans (skeletonLT X (n + 1)).carrier (skeletonLT X n) C
    ↪ (skeletonLT_mono
      le_self_add) ?_ (Subcomplex.base_subset (skeletonLT X ↑n)) n_ih
    exact HEP'_skeleton X n
```

The base case is the lemma `HEP_self`, which we proved at the very beginning, just after defining the HEP. Getting there takes the rewrite that the base space is the same as the (-1) -skeleton. For the induction step we combine the induction hypothesis with the HEP for two consecutive skeletons via the transitivity lemma `HEP_trans`.

The corresponding statement for `skeleton` then follows from the one for `skeletonLT`, since `skeleton X m` in `skeletonLT X (m+1)`.

```
lemma HEP_skel_base (m : ℕ) : HEP' (skeleton X m) C := by
  rw [skeleton.congr_simp X m m rfl]
  exact HEP_skelLT_base X (m + 1)
```

It remains to show that for a finite dimensional complex there is an $m \in \mathbb{N} \cup \{-1\}$ with $X = \text{skeletonLT } X \ m$. This is the content of the following lemma.

```
lemma RelCWFinite_eq_SkelLT (hX : FiniteDimensional X) : ∃ (m : ℕ), X =
  ↪ skeletonLT X m := by ...
```

The idea of the proof is simple. Being finite dimensional means that the index types of the cells are empty from some dimension m on. The complex itself can be written as the union of the base space with all open cells, and in `RelCW_eq_UnionSkelLTCell` we show that it can equally be written as the union of `skeletonLT X m` with the open cells of dimension at least m . For the m given by finite dimensionality the second part of this union is empty, which gives the claim.

Putting the two steps together, the final proof is short.

```
lemma HEP_RelCWFinite (hX : FiniteDimensional X) : HEP' X C := by
  obtain ⟨m, hm⟩ := RelCWFinite_eq_SkelLT X hX
  rw [hm]
  exact HEP_skelLT_base X m
```

This completes the three steps announced at the beginning: the *HEP* for a ball and a cube relative to their boundary, for two consecutive skeleta, and finally for a finite dimensional relative CW complex relative to its base space.

ToDo Ausblick: In the previous section, where we proved the *HEP* for consecutive skeletons, I used the lemma `retractioncriterionclosed`, where the retraction map is defined on the Type

ToDo SkelLT und skel erklären

5 Methodes and Usage of AI

The use of AI as an aid for this thesis was agreed upon in advance with my supervisor, Floris van Doorn, on the condition that it be made transparent here and that any code not written by myself be marked as such in the formalisation. My methods, and with them the usage of AI, differed between the formalisation project and the writing of the text. In both cases, whenever AI was used, it was the latest version of Anthropic’s Claude Opus model (as of August 2026: Opus 5).

All submitted Lean code is written by myself, except a short definition together with a related lemma in the file `HEP_skeleton.lean`. These approximately twenty lines marked accordingly in the formalisation. This lemma was a tool that could have been a part of `mathlib`. It is not specific to the constructions of this project.

The rest of the formalisation is my own work, but it would in some parts look different without the use of AI. I frequently asked Claude when a proof got stuck or an error message was difficult to interpret. One example is the tactic `subst`, which simplified the proof of `SkeletonProjectionIsQuotientMap` a lot and later allowed me to shorten several earlier drafts by a noticeable number of lines. Another is the observation that consecutive `exact` statements can be combined into a single `exacts [...]`.

For larger design questions I found AI less helpful. The pushout construction for the retraction of consecutive `skelta` used in the lecture notes, for instance, did not match the formalisation of CW-complexes in `mathlib` that I wanted to work with. I asked Claude for help at this point. Instead of working within the subset setting of the formalised CW-complexes, which is the solution presented earlier, it suggested a longer and more technical way that did not seem realisable to me. The biweekly meetings with Floris van Doorn were the place where such decisions were actually resolved.

I wrote the text of this thesis in full myself in a first step. In a second pass I went through it paragraph by paragraph and improved the language, along with some inaccuracies, with the help of Claude. Every reformulation was reviewed before I adopted it, and I discarded those that shifted the meaning of what I had written. AI was also used to set up the configuration for typesetting Lean code in this $\text{\LaTeX}$ document.

6 Ausblick

References

- [] URL: <https://www.math.uni-bielefeld.de/~fw/> (visited on 08/14/2026).
- [] *Mathlib Statistics*. URL: https://leanprover-community.github.io/mathlib_stats.html (visited on 07/02/2026).
- [Avi+24] Jeremy Avigad et al. *Theorem Proving in Lean 4*. https://leanprover.github.io/theorem_proving_in_lean4/. Zugriff am 2. Juli 2026. 2024.
- [FRO] Lean FRO. *FRO - About us*. URL: <https://lean-lang.org/fro/about/> (visited on 07/02/2026).
- [Hat02] Allen Hatcher. *Algebraic topology*. Cambridge: Cambridge University Press, 2002, pp. xii+544. ISBN: 0-521-79540-0.
- [Kon25] Alex Kontorovich. *The Shape of Math To Come*. 2025. arXiv: 2510.15924 [math.HO]. URL: <https://arxiv.org/abs/2510.15924>.
- [MLS25] Alex Meiburg, Leonardo A. Lessa, and Rodolfo R. Soldati. *A Formalization of the Generalized Quantum Stein’s Lemma in Lean*. 2025. arXiv: 2510.08672 [quant-ph]. URL: <https://arxiv.org/abs/2510.08672>.
- [Sch24] Hannah Scholz. “Formalisation of CW-complexes”. Bachelor’s Thesis Mathematics. Rheinischen Friedrich-Wilhelms-Universität Bonn, 2024. URL: <https://florisvandoorn.com/theses/HannahScholz.pdf>.
- [Str68] Arne Strøm. “Note on Cofibrations II”. In: *Mathematica Scandinavica* 22 (1968), pp. 130–142.
- [Whi18] J. H. C. Whitehead. “Combinatorial homotopy. I”. In: *Bulletin (new series) of the American Mathematical Society* 55.3 (2018), pp. 213–245. ISSN: 0273-0979.